

Principles of climate models

Video transcript

To create a climate model, the Earth is typically divided into a 100 square kilometer grid. Each box contains climate information relevant to its location. Depending on the level of detail we need in our forecast and how far ahead we want to go in time, the size of the boxes can be made smaller to include more local detail or bigger to reduce the number of calculations required for a long period forecast. In this example, our box sits just on the surface. Inside we can see that it contains information about the land, information on the vegetation and if it's an urban area or not.

All these aspects contribute to the climate system and can affect the surrounding area. The further into the future we want to predict, the more elements of the Earth's climate system we need to include in our model. For example, the global oceans and sea ice, interactive vegetation, which can change with the climate, atmospheric chemistry, man-made pollution sources, and a model of how carbon is transferred between the atmosphere, oceans and the land surface.

All of these calculations and complexities mean that many billions of computations have to be carried out in order to produce a projection of the climate into the future. Every element of the model must be written down in computer code and fed into a large supercomputer. Once provided with some initial conditions, the model can then run forwards in time to produce climate projections for whatever timescale we want to study.

Using climate models, we are able to get an understanding of the past, present, and future climate. It allows us to determine how natural and man-made influences impact the climate, and can help us create solutions to combat climate change in the future.